



Modelling Financial Markets

Chapter 6: Multivariate Volatility and Dependence



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ BAC: Bank of America Corporation (NYSE ticker)
- ▣ BEKK: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNP: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ CCC: Constant Conditional Correlation (model)
- ▣ CDO: Collateralised Debt Obligation
- ▣ CML: Canonical Maximum Likelihood
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CPI: Consumer Price Index
- ▣ DAX: Deutscher Aktienindex (the German stock index (Frankfurt Stock Exchange))
- ▣ DBK: Deutsche Bank AG (Xetra ticker)
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DCC-GARCH: Dynamic Conditional Correlation GARCH (model)
- ▣ DCC-NL: Dynamic Conditional Correlation with NonLinear shrinkage
- ▣ EU: European Union
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FRED: Federal Reserve Economic Data
- ▣ FRM: Financial Risk Meter
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GIRF: Generalised Impulse Response Function

Acronyms used in this material (2/2)

- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- IFM: Inference Functions for Margins
- IIT: Illinois Institute of Technology
- JPM: JPMorgan Chase & Co. (NYSE ticker)
- PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- TTF: Title Transfer Facility (Dutch natural gas hub)
- TVP-VAR: Time-Varying Parameter Vector AutoRegression
- US: United States
- VEC: VECtorised multivariate GARCH model (vech of the covariance matrix)
- WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Why Correlations Matter

Rolling and EWMA Covariances

Multivariate GARCH and DCC

Crisis Correlations and the Forbes–Rigobon Bias

Copulas and Tail Dependence

Applications: Europe, Romania and Banks

Case Study from Our Research: Energy Prices and EU Stock Markets

AI for Scientific Discovery

Summary

References

Why Dependence Matters

Three reasons

- **Diversification:** portfolio risk depends on how assets move together
 - ▶ **Markowitz (1952):** variance of a portfolio is a quadratic form in the covariance matrix
- **Hedging:** the hedge ratio is a conditional regression slope
 - ▶ If the correlation changes, yesterday's hedge is wrong today
- **Contagion:** shocks spread across markets and institutions
 - ▶ Correlations tend to rise exactly when diversification is needed most

What we do in this chapter

- Estimate time-varying covariances: rolling windows, EWMA, multivariate GARCH, DCC
- Separate true contagion from the volatility bias (Forbes–Rigobon)
- Go beyond correlation: copulas and tail dependence
- Apply everything to stocks and bonds, Europe and Romania, Bitcoin and banks

Recap from Chapter 4: Correlation in Portfolio Risk

- Portfolio variance $\sigma_p^2 = w^\top \Sigma w$; with two assets the correlation enters only the cross term $2w(1-w)\rho\sigma_1\sigma_2$
- SPY (S&P 500 exchange-traded fund) and TLT (20+ year US Treasury bond fund), 60/40 portfolio
 - ▶ Volatility 10.4% with the 2002–2021 correlation (-0.40), 13.2% with the 2022–2026 one (0.11), same asset volatilities
- Minimum-variance hedge ratio $h^* = \rho\sigma_S/\sigma_F$ (the slope of r_S on r_F), effectiveness ρ^2 : a dynamic hedge needs a dynamic ρ_t (Section 3)
- What is new in this chapter
 - ▶ Σ_t moves: how to estimate it, and how to measure the sampling uncertainty of every dependence estimate

Contagion: When Correlations Jump



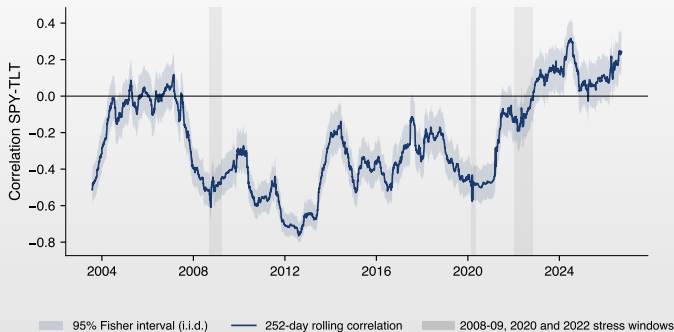
Lehman Brothers headquarters, New York, 15 September 2008, the day of the bankruptcy filing

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia

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- **Interdependence:** markets are linked in all states of the world
 - ▶ Trade, common investors, common macro factors
- **Contagion** ([Forbes & Rigobon \(2002\)](#)): a *significant increase* in cross-market linkages after a shock
 - ▶ A high correlation during a crisis is not enough; it must be higher than in calm times
- Section 4 shows why a naive comparison of correlations overstates contagion

Stocks and Bonds: A Correlation That Changed Sign



- ▣ Correlation of daily SPY and TLT returns on a 252-day window, with a 95% Fisher interval that assumes i.i.d. data
- ▣ Grey bands: autumn 2008 – March 2009, February–April 2020 and January–October 2022

Interpreting the Stock–Bond Correlation

- Minimum -0.77 (8 August 2012), maximum 0.32 (1 July 2024), last value 0.24
- Share of days with a positive rolling correlation: 6% before 2022, 81% since January 2022
- The sign depends on what drives bond yields
 - ▶ Growth shocks: stocks fall, yields fall, bonds rise (negative correlation)
 - ▶ Inflation shocks: stocks and bonds fall together (positive correlation), as in 2022
- The Fisher band is too narrow here: it ignores fat tails and volatility clustering (Section 2 gives the corrected band)

Section Recap

- ▣ Portfolio risk is $w^\top \Sigma w$: the off-diagonal terms often dominate for large portfolios
- ▣ The minimum-variance hedge ratio is $\rho \sigma_S / \sigma_F$; its effectiveness is ρ^2
- ▣ Contagion means a significant *increase* of linkages, not just high co-movement
- ▣ The SPY–TLT correlation moved from mostly negative to mostly positive in 2022: correlations are not constants

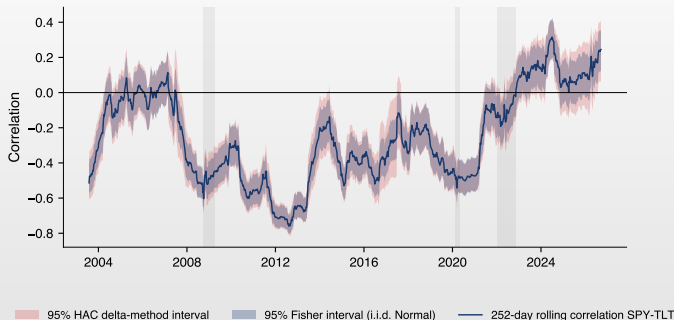
Sample Covariance and Its Precision

- ▣ Returns $r_t \in \mathbb{R}^N$, $t = 1, \dots, T$; sample covariance $\hat{\Sigma} = \frac{1}{T-1} \sum_t (r_t - \bar{r})(r_t - \bar{r})^\top$
- ▣ Pearson correlation: $\hat{\rho}_{ij} = \hat{\Sigma}_{ij} / \sqrt{\hat{\Sigma}_{ii} \hat{\Sigma}_{jj}} \in [-1, 1]$
- ▣ Fisher transformation: $z = \text{artanh } \hat{\rho} \approx N(\text{artanh } \rho, 1/(T-3))$ for i.i.d. Normal data
 - ▶ 95% interval: $\tanh(z \pm 1.96/\sqrt{T-3})$
 - ▶ With $T = 252$ and $\rho = 0.5$ the half-width is about 0.09: a one-year correlation is noisy
- ▣ N assets: $N(N+1)/2$ distinct elements; with $N = 500$ this is 125 250 numbers

Inference on a Correlation under Volatility Clustering

- ▣ $\hat{\rho} = g(\hat{m})$, with $\hat{m}$ the sample mean of $m_t = (x_t, y_t, x_t^2, y_t^2, x_t y_t)$
- ▣ Delta method: $\sqrt{T}(\hat{\rho} - \rho) \rightarrow N(0, \nabla g^\top \Omega \nabla g)$, Ω the long-run covariance of m_t ; requires stationarity, finite fourth moments of (x_t, y_t) and weak dependence (mixing)
 - ▶ $\hat{\Omega}$: HAC estimator with Bartlett kernel and L lags (Newey & West (1987))
 - ▶ Equivalent form: $\hat{\rho} - \rho \approx \frac{1}{T} \sum_t \psi_t$, $\psi_t = \tilde{x}_t \tilde{y}_t - \frac{\rho}{2}(\tilde{x}_t^2 + \tilde{y}_t^2)$, $\tilde{x}, \tilde{y}$ standardised
- ▣ i.i.d. Normal pairs: $\text{Var}(\psi_t) = (1 - \rho^2)^2$, the Fisher variance; fat tails typically raise $\text{Var}(\psi_t)$ and GARCH can make ψ_t autocorrelated; with infinite fourth moments the Normal limit fails
- ▣ Interval on the Fisher scale: $\tanh(\text{artanh } \hat{\rho} \pm 1.96 \hat{\text{se}} / (1 - \hat{\rho}^2))$
- ▣ SPY–TLT daily, 2002–2021 ($T = 4891$, $\hat{\rho} = -0.403$): se 0.0120 (Fisher), 0.0217 (fourth moments, $L = 0$), 0.0227 (HAC, $L = 9$)

Fisher versus HAC Intervals: SPY and TLT



- 252-day rolling correlation of daily returns, every 5th day, with the 95% Fisher interval and the 95% HAC delta-method interval (Bartlett, $L = 4$)

Interpreting the HAC Intervals

- The HAC interval is wider in almost every window: median width ratio 1.24, 90% quantile 1.54, maximum 2.02
- Windows with a correlation significantly different from zero: 73% by Fisher, 65% by HAC; the verdict flips in 8.2% of the windows
- Last window (17 September 2026): $\hat{\rho} = 0.25$, Fisher [0.13, 0.36], HAC [0.07, 0.41]
- Whole periods: the HAC standard error is 1.9 times Fisher in 2002–2021 and 1.4 times since 2022; fourth moments alone give 1.8 before 2022, so fat tails matter more than autocorrelation
- Rule for daily returns: report delta-method HAC intervals for correlations

Rolling-Window Estimators

- ▣ Estimate on the last m observations: $\hat{\Sigma}_t = \frac{1}{m} \sum_{s=t-m}^{t-1} r_s r_s^\top$ (mean set to zero for daily data)
- ▣ Trade-off in m
 - ▶ Short window: reacts fast, but noisy
 - ▶ Long window: stable, but slow to detect a regime change
- ▣ **Ghost effect**: an extreme day enters with weight $1/m$ and leaves abruptly m days later
- ▣ Always positive semidefinite ($v^\top \hat{\Sigma}_t v = \frac{1}{m} \sum_s (v^\top r_s)^2 \geq 0$); singular if $m < N$; positive definite only if the m return vectors span $\mathbb{R}^N$

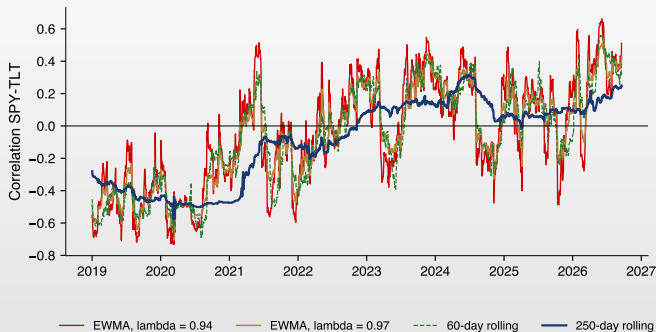
EWMA: The RiskMetrics Recursion

- Exponentially decaying weights:
 $\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda) r_{t-1} r_{t-1}^\top$
- Equivalent: $\Sigma_t = (1 - \lambda) \sum_{k \geq 1} \lambda^{k-1} r_{t-k} r_{t-k}^\top$
- J.P. Morgan/Reuters (1996): $\lambda = 0.94$ for daily and $\lambda = 0.97$ for monthly data
- Half-life of a shock: $\ln 0.5 / \ln \lambda = 11.2$ periods ($\lambda = 0.94$), 22.8 periods ($\lambda = 0.97$); a period is a day for daily data and a month for monthly data

Properties

- Positive semidefinite: a weighted sum of outer products
- No ghost effect: weights decay smoothly
- Same λ for all variances and covariances: very restrictive
- It is an integrated GARCH(1,1) without a constant: no mean reversion (Chapter 5)

EWMA versus Rolling Windows, SPY-TLT since 2019



- Two EWMA estimators ($\lambda = 0.94$ and 0.97) and two rolling windows (60 and 250 days) on the same data

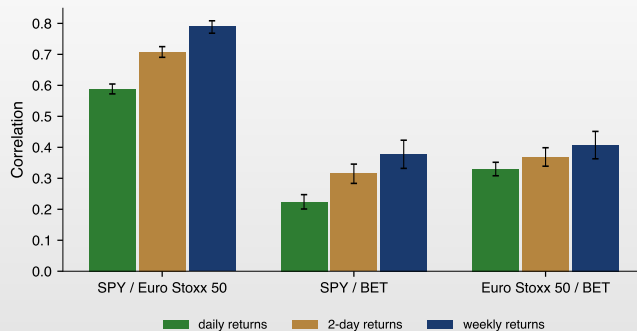
Interpreting the EWMA Comparison

- Standard deviation of the daily change in the estimate: 0.060 (EWMA 0.94), 0.032 (EWMA 0.97), 0.0088 (250-day window)
- EWMA with $\lambda = 0.94$ uses an effective sample of about $(1 + \lambda)/(1 - \lambda) \approx 32$ days: fast but noisy
- Persistently positive (20 days in a row) from 10 May 2022 with EWMA 0.94, but only from 8 December 2022 with the 250-day window
- Here λ and m are fixed in advance; DCC (Section 3) estimates its own dynamic parameters (a, b) from the data

Asynchronous Trading: A Trap for Daily Correlations

- Closing times differ: Bucharest and Frankfurt close hours before New York
 - ▶ US news after the European close enters European prices only the next day
- Consequence: same-day correlation is biased towards zero, and the lagged correlation is positive
- Remedies
 - ▶ Lower frequency: 2-day or weekly returns on common trading days
 - ▶ Add the lead-lag term: $\rho_0 + \rho_{+1}$ approximates the synchronous correlation
- Rule used throughout: join the *prices* on common days first, then compute returns

Daily, 2-Day and Weekly Correlations, 2000–2026



□ SPY, Euro Stoxx 50 and BET on common trading days; error bars: 95% Fisher intervals

Interpreting the Frequency Effect

- SPY–Euro Stoxx 50: 0.59 daily, 0.71 on 2-day returns, 0.79 weekly
- Lagged daily correlation (SPY yesterday, Euro Stoxx 50 today): 0.20; for BET: 0.17
- SPY–BET: 0.22 daily versus 0.38 weekly; Euro Stoxx 50–BET (almost synchronous closes): 0.33 versus 0.41
- The gap is largest across time zones: part of the “low” US–Europe daily correlation is a measurement artefact
- Weekly data are cleaner but give 1 367 instead of 6 446 observations: wider intervals

Section Recap

- Rolling windows: simple, but the window length is arbitrary and extreme days create ghost effects
- EWMA: $\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda) r_{t-1} r_{t-1}^\top$, positive semidefinite, one decay for all elements, no mean reversion
- Sampling error of a correlation is large; for SPY-TLT Fisher intervals understate it; delta-method HAC intervals are asymptotically valid under finite fourth moments and weak dependence
- Across time zones use 2-day or weekly returns, or add lead-lag terms

From Univariate to Multivariate Volatility

- Chapter 5: $r_t = \sigma_t z_t$ with $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$
- Now: $r_t = H_t^{1/2} z_t$, H_t known at $t-1$ ($\mathcal{F}_{t-1}$ -measurable), $\mathbb{E}(z_t | \mathcal{F}_{t-1}) = 0$, $\text{Var}(z_t | \mathcal{F}_{t-1}) = I_N$, so $\text{Var}(r_t | \mathcal{F}_{t-1}) = H_t$
- Any model for H_t must deliver
 - ▶ Positive definiteness at every t
 - ▶ Few parameters, so that estimation is feasible for many assets
 - ▶ Interpretable dynamics: volatility spillovers, time-varying correlation
- Gaussian quasi-log-likelihood: $\ell = -\frac{1}{2} \sum_t (N \ln 2\pi + \ln |H_t| + r_t^\top H_t^{-1} r_t)$

The VEC and BEKK Models

- **VEC** (Bollerslev, Engle & Wooldridge (1988)): every element of H_t depends on all squares and cross-products
 - ▶ $\text{vech}(H_t) = c + A \text{vech}(r_{t-1} r_{t-1}^\top) + B \text{vech}(H_{t-1})$
 - ▶ Positive definiteness is not guaranteed
- **BEKK** (Engle & Kroner (1995)): positive definite by construction
 - ▶ $H_t = CC^\top + A^\top r_{t-1} r_{t-1}^\top A + B^\top H_{t-1} B$
 - ▶ Parameters are hard to interpret; the likelihood is flat in high dimension

Number of parameters

N	VEC	full BEKK	DCC
2	21	11	$6 + 2$
5	465	65	$15 + 2$
10	6 105	255	$30 + 2$
50	3 252 525	6 275	$150 + 2$

- VEC: $N(N+1)/2 \cdot [1 + N(N+1)]$; BEKK: $N(N+1)/2 + 2N^2$
- DCC: $3N$ univariate GARCH parameters plus 2 (correlation targeted)

Constant and Dynamic Conditional Correlation

- Decomposition: $H_t = D_t R_t D_t$, $D_t = \text{diag}(\sigma_{1t}, \dots, \sigma_{Nt})$ from univariate GARCH
- CCC (Bollerslev (1990)): $R_t = R$ constant; standardised residuals $\varepsilon_{it} = r_{it}/\sigma_{it}$
- DCC (Engle (2002)): a GARCH-type recursion for a proxy matrix Q_t
 - ▶ $Q_t = (1 - a - b) \bar{Q} + a \varepsilon_{t-1} \varepsilon_{t-1}^\top + b Q_{t-1}$, $a, b \geq 0$, $a + b < 1$
 - ▶ $R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$
 - ▶ $\bar{Q}$: sample covariance of ε_t (correlation targeting)
- Two parameters govern all $N(N-1)/2$ correlations: parsimonious, but every pair has the same persistence

Robert Engle and the DCC Model



Robert F. Engle at the Western Economic Association
International conference, Santiago, 2017

Photo: [Econterms \(2017\)](#); CC BY-SA 4.0; [Wikimedia](#)

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- Nobel Memorial Prize in Economic Sciences 2003 for ARCH models (Chapter 5)
- [Engle \(2002\)](#): “Dynamic Conditional Correlation”
 - ▶ Keeps the flexibility of univariate GARCH for each variance
 - ▶ Adds a simple, estimable law of motion for correlations
- Refinement: the corrected DCC of [Aielli \(2013\)](#), which makes $\bar{Q}$ a consistent target

Two-Step Estimation

- The Gaussian log-likelihood splits into a volatility part and a correlation part

$$\ell = \underbrace{-\frac{1}{2} \sum_t (N \ln 2\pi + 2 \ln |D_t| + r_t^\top D_t^{-2} r_t)}_{\ell_V} \underbrace{-\frac{1}{2} \sum_t (\ln |R_t| + \varepsilon_t^\top R_t^{-1} \varepsilon_t - \varepsilon_t^\top \varepsilon_t)}_{\ell_C}$$

- Step 1: fit N univariate GARCH(1,1) models; keep σ_{it} and ε_{it}
- Step 2: fix $\bar{Q}$ at the sample covariance of ε_t , maximise ℓ_C over (a, b)
- Inference
 - ▶ Quasi-likelihood: Normality of z_t is not needed if conditional means and covariances are correctly specified; but sample-covariance targeting makes the standard DCC estimator inconsistent in general, the corrected DCC is not (Aielli (2013))
 - ▶ Standard errors from step 2 alone treat the GARCH estimates as known: the next slide gives the correction
 - ▶ Test CCC against DCC: $H_0 : a = 0$; b is not identified under H_0 , so the χ^2 reference is only indicative

Two-Step Estimators: The Correct Covariance

- Stack the estimating equations: $\sum_t s_{1t}(\theta_1) = 0$ (step 1, GARCH) and $\sum_t s_{2t}(\theta_1, \theta_2) = 0$ (step 2, correlation)
- $A_{jk} = \mathbb{E}[\partial s_{jt} / \partial \theta_k^\top]$: A_{21} measures how the correlation score reacts to the GARCH parameters
- Newey & McFadden (1994), Section 6: $\sqrt{T}(\hat{\theta}_2 - \theta_2) \rightarrow N(0, V_2)$ with
 - ▶ $V_2 = A_{22}^{-1} \mathbb{E}[(s_{2t} - A_{21} A_{11}^{-1} s_{1t})(s_{2t} - A_{21} A_{11}^{-1} s_{1t})^\top] A_{22}^{-1\top}$
- Step-2 standard errors keep only s_{2t} : they drop $A_{21} A_{11}^{-1} s_{1t}$, the GARCH estimation error carried into step 2
- Conditions: with correct conditional moments the scores are martingale differences, so the middle term is their variance; otherwise use its long-run (HAC) version
 - ▶ With targeting, $\bar{Q}$ is estimated too: its moment equations belong to step 1 (θ_1 includes $\text{vech } \bar{Q}$)
- Engle & Sheppard (2001) derive V_2 for DCC; a simple alternative is to bootstrap the *whole* two-step procedure, re-estimating GARCH, $\bar{Q}$ and (a, b) on every path

DCC Estimates for SPY and TLT

Step 1: GARCH(1,1), returns in %

	ω	α	β	$\alpha + \beta$
SPY	0.0283	0.127	0.848	0.975
TLT	0.0057	0.049	0.943	0.993

Step 2: DCC(1,1)

	estimate	std. error
a	0.0522	0.0059
b	0.9335	0.0084
$a + b$	0.9857	
$\tilde{Q}_{12}$	-0.248	

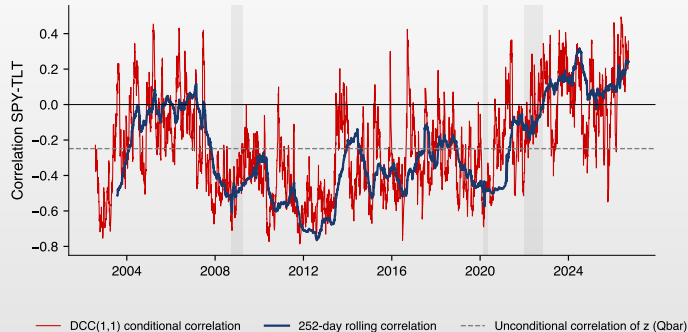
- Daily data 31 July 2002 – 18 September 2026, $T = 6\,073$
- Correlation persistence $a + b = 0.986$: half-life of a shock to $Q_t \approx 48$ days, approximate because $\mathbb{E}_{t-1}[\varepsilon_t \varepsilon_t^\top] = R_t \neq Q_t$
- Likelihood ratio DCC versus CCC: $2(\ell_C^{DCC} - \ell_C^{CCC}) = 533.4$
- Under H_0 : CCC, b is not identified: parametric bootstrap of both steps under CCC ($B = 199$), largest null statistic 15.7, $p = 0.005$: CCC is rejected
- TLT volatility is more persistent (β larger) but reacts less to news (α smaller) than SPY volatility

Two-Step Standard Errors for SPY and TLT

	$\hat{a}$	$\hat{b}$	$\hat{a} + \hat{b}$
estimate	0.0522	0.9335	0.9857
se, step-2 Hessian	0.0059	0.0084	–
se, bootstrap of step 2 only	0.0051	0.0077	0.0038
se, bootstrap of both steps	0.0052	0.0077	0.0040

- Parametric residual bootstrap, $B = 199$: DCC innovations simulated with $(\hat{a}, \hat{b}, \bar{Q})$ from resampled decorrelated residuals
- Step 2 only: re-estimate (a, b) on the simulated innovations; both steps: rebuild returns with the fitted GARCH, re-estimate GARCH, then DCC (same innovations)
- Step 1 raises the se by only 2% (a) and 1% (b): $T = 6\,073$ gives precise GARCH estimates; the step-2 Hessian understates $\text{se}(\hat{b})$ by -8% because non-Normal innovations call for a sandwich

DCC Correlation of SPY and TLT, 2002–2026



- Conditional correlation $\rho_{12,t}$ from the DCC(1,1) model, the 252-day rolling correlation and the target $\bar{Q}_{12}$

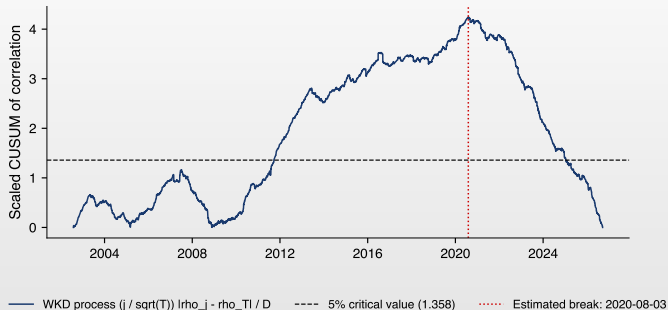
Interpreting the DCC Path

- ▣ Range: from -0.79 (2 November 2011) to 0.49 (21 May 2026); last value 0.36
- ▣ Average DCC correlation: -0.54 from 15 September to 31 December 2008, -0.56 in 2012, 0.00 in 2022
- ▣ DCC reacts within days, like EWMA, but its speed (a) and memory (b) are estimated, not imposed
- ▣ Mean reversion to $\bar{Q}_{12} = -0.25$ is a weakness when the regime changes
 - ▶ Since January 2022 the path is above the target on 92% of days (36% before): a single long-run correlation is doubtful

Testing Constant Correlation at an Unknown Date

- Tse (2000): a Lagrange multiplier test of constant conditional correlation, computed from the CCC fit alone
- Picking a break date from a chart and then testing it with a fixed-date Fisher z is oversized: the date is itself a function of the data
- Wied, Krämer & Dehling (2012): $Q_T = \max_{2 \leq j \leq T} \frac{j}{\sqrt{T} \hat{D}} |\hat{\rho}_j - \hat{\rho}_T|$, $\hat{\rho}_j$ the correlation of the first j observations
 - ▶ $\hat{D}^2$: the delta-method long-run variance of $\hat{\rho}$ (Section 2), Bartlett kernel with bandwidth $\lfloor T^{1/4} \rfloor$
 - ▶ Under H_0 (constant correlation, means and variances): $Q_T \rightarrow \sup_{0 \leq s \leq 1} |B(s)|$, B a Brownian bridge; 5% critical value 1.358
 - ▶ Constant variances are required, so we apply it to GARCH-standardised residuals; break date estimate $\hat{j} = \arg \max$

Dating the SPY–TLT Correlation Break



- Wied–Krämer–Dehling process on the GARCH(1,1)-standardised residuals of SPY and TLT, $T = 6\,073$, bandwidth 8; dashed: 5% critical value

Interpreting the Break Test

- ▣ $Q_T = 4.25$ against the 5% critical value 1.358: constant correlation of the residuals is rejected ($p < 0.001$)
- ▣ Estimated break: 3 August 2020, not January 2022; residual correlation -0.36 before and 0.07 after
- ▣ Binary segmentation: before the break $Q = 3.01$ (second break 9 July 2007); after it $Q = 1.56$, $p = 0.015$: more than one break
- ▣ Caveat: the limit law is proved for observed series; on estimated GARCH residuals the p -values are approximate
- ▣ For DCC: a single target $\bar{Q}$ over 2002–2026 averages at least two regimes

Application: Minimum-Variance Weights with DCC



- Weight in SPY of the daily minimum-variance SPY-TLT portfolio: $w_t = (\sigma_{2t}^2 - \sigma_{12,t}) / (\sigma_{1t}^2 + \sigma_{2t}^2 - 2\sigma_{12,t})$, truncated to $[0, 1]$
- Same GARCH volatilities; correlation from DCC or constant (-0.30)

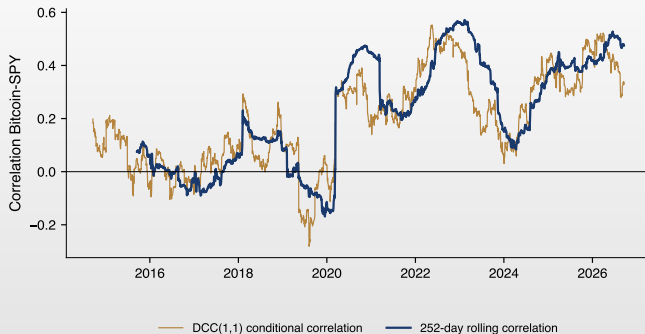
Interpreting the Portfolio Application

- ▣ Annualised volatility (%): 60/40 fixed 10.99; GARCH + constant correlation 9.10; GARCH + DCC 9.06
- ▣ Most of the gain comes from GARCH volatilities and minimum-variance weights; the DCC correlation adds only a small extra reduction
- ▣ Average absolute weight difference DCC minus constant: 0.019; largest: 0.22 (8 June 2026)
- ▣ Caveat: parameters are estimated on the full sample, so this is an in-sample illustration
 - ▶ A fair comparison re-estimates the models on expanding windows and adds transaction costs (Chapter 8)

Dynamic Hedging Out of Sample: BET with Euro Stoxx 50

- Daily BET and Euro Stoxx 50 (almost synchronous closes) from 5 January 2010; estimation to end-2019 ($n = 2\,491$), evaluation 2020–2026 ($n = 1\,663$) with frozen parameters
- Static hedge: slope of BET on Euro Stoxx 50 in the estimation window, $h = 0.337$
- Dynamic hedge: $h_t = \rho_t \sigma_{S,t} / \sigma_{F,t}$ from GARCH(1,1) and DCC(1,1) ($\hat{a} = 0.009$, $\hat{b} = 0.964$): mean 0.29, range 0.13 to 0.84
- Variance removed in 2020–2026: static 24.7%, DCC 26.2%; the best constant hedge chosen ex post ($h = 0.41$): 25.6%
- Diebold & Mariano (1995) test on squared hedged returns (static minus DCC), HAC: $t = 1.29$, one-sided $p = 0.098$: a small gain, not significant at 5%

Bitcoin and US Equities: DCC Correlation



- Bitcoin and SPY on common trading days since 18 September 2014 ($T = 3018$); weekend Bitcoin moves enter the Monday return

Interpreting the Bitcoin–Equity Correlation

- Average DCC correlation: 0.04 before 2020, 0.31 since January 2020; peak 0.55 on 19 May 2022
- Very persistent dynamics: $a = 0.015$, $b = 0.982$, $a + b = 0.9968$; likelihood ratio 105.6, bootstrap p under CCC = 0.005: CCC rejected
- From diversifier to risk asset: since 2020 Bitcoin co-moves with equities much more than before
 - ▶ [Pele et al. \(2023\)](#) ask whether cryptos have become an asset class of their own; the rising correlation is part of the answer
- The full-sample correlation (0.24) hides two different regimes

Section Recap

- ▣ $H_t = D_t R_t D_t$ separates volatilities (univariate GARCH) from correlations
- ▣ VEC and BEKK are general but explode in parameters; DCC needs $3N + 2$
- ▣ Two-step quasi-likelihood needs correctly specified conditional moments, not Normality; standard DCC targeting is inconsistent in general (corrected DCC fixes it); step-2 standard errors miss the GARCH error: use Newey–McFadden or bootstrap both steps
- ▣ An unknown-date test dates the SPY–TLT residual break to 3 August 2020; a static hedge loses little to DCC out of sample (BET)
- ▣ SPY–TLT: $a + b = 0.986$, CCC rejected by a bootstrap under H_0 , sign change after 2022; Bitcoin–SPY: correlation up from 0.04 to 0.31

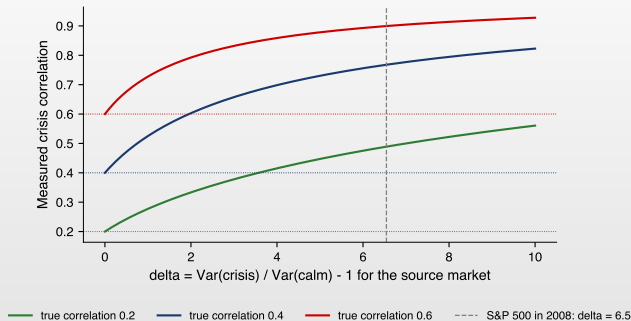
Do Correlations Rise in Crises?

- ▣ Common finding: correlations measured during crises are higher than in calm periods
- ▣ Two very different explanations
 - ▶ **Contagion**: the transmission mechanism itself changes
 - ▶ **Volatility bias**: the mechanism is unchanged, but a more volatile source market mechanically raises the correlation
- ▣ **Forbes & Rigobon (2002)**: after correcting the bias, most of the 1987, 1994 and 1997 “contagion” became interdependence
- ▣ **Longin & Solnik (2001)**, **Ang & Chen (2002)**: correlations rise in bear markets, not in bull markets: an asymmetry the Normal distribution cannot produce

The Heteroskedasticity Bias: Derivation

- Model: $y_t = \alpha + \beta x_t + \varepsilon_t$, $\text{Cov}(x_t, \varepsilon_t) = 0$, β and σ_ε^2 constant
- Correlation: $\rho^2 = \frac{\beta^2 \sigma_x^2}{\beta^2 \sigma_x^2 + \sigma_\varepsilon^2} = \left(1 + \frac{\sigma_\varepsilon^2}{\beta^2 \sigma_x^2}\right)^{-1}$
- Crisis: $\sigma_{x,c}^2 = (1 + \delta) \sigma_x^2$ with $\delta > 0$; nothing else changes
- Then $\rho_c = \rho \sqrt{\frac{1 + \delta}{1 + \delta \rho^2}}$, so $|\rho_c| > |\rho|$ for $0 < |\rho| < 1$ ($\rho_c > \rho$ when $\rho > 0$): the measured correlation rises in absolute value although β is the same
- Inverting: $\rho^* = \frac{\rho_c}{\sqrt{1 + \delta(1 - \rho_c^2)}}$ is the crisis correlation adjusted for the volatility increase

How Much Can Volatility Alone Raise a Correlation?



- Measured crisis correlation ρ_c as a function of δ , for three true correlations that do not change
- Vertical line: the variance of 2-day S&P 500 returns was 7.5 times its calm-year level in the 2008 crisis window ($\delta = 6.54$)

The Forbes–Rigobon Test

□ Procedure

- ▶ Choose the source market x , a calm window and a crisis window
- ▶ Compute $\delta = \hat{\sigma}_{x,c}^2 / \hat{\sigma}_{x,calm}^2 - 1$ and the adjusted ρ^*
- ▶ Test $H_0 : \rho^* \leq \rho_{calm}$ with Fisher z statistics (the published convention)
- ▶ Caveat: with δ fixed, $\text{Var}(\text{artanh } \rho^*) \approx 1/[n_c(1 + \delta(1 - \rho_c^2))] < 1/n_c$, so this test is conservative; an estimated δ and overlapping 2-day returns (n halved, an approximation) need a joint delta method or a block bootstrap
- Use 2-day returns to reduce the time-zone problem (as in Forbes & Rigobon (2002))

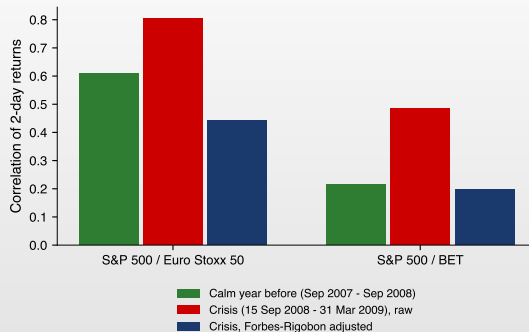
Assumptions to remember

- No omitted common shock: a global factor hitting both markets also raises σ_x
- No feedback from y to x (no endogeneity)
- Choice of windows matters: results can flip with a few weeks
- The correction can go too far if the true mechanism did change

Worked Example: The 2008 Crisis Step by Step

- Source market: S&P 500; calm year: 14 September 2007 – 12 September 2008 ($n = 242$ 2-day returns); crisis: 15 September 2008 – 31 March 2009 ($n = 130$)
- Variance ratio: $1 + \delta = 7.54$, so $\delta = 6.54$
- Euro Stoxx 50: $\rho_{calm} = 0.610$, $\rho_c = 0.804$
 - ▶ $\rho^* = 0.804 / \sqrt{1 + 6.54(1 - 0.804^2)} = 0.442$
 - ▶ One-sided p -value: 0.005 with ρ_c , 0.932 with ρ^*
- BET: $\rho_{calm} = 0.216$, $\rho_c = 0.485$, $\rho^* = 0.198$
 - ▶ One-sided p -value: 0.024 with ρ_c , 0.548 with ρ^*

2008: Raw versus Adjusted Crisis Correlations



- Correlation of 2-day returns with the S&P 500 in the calm year, in the crisis (raw) and after the Forbes–Rigobon adjustment

Interpreting the 2008 Test

- ▣ Raw correlations jump: Euro Stoxx 50 from 0.61 to 0.80, BET from 0.22 to 0.48; both increases significant at 5%
- ▣ After the adjustment neither increase survives: the 2008 co-movement is consistent with “no contagion, only interdependence”
- ▣ Why interpret these results cautiously?
 - ▶ 2008 was a global shock that hit all markets at once: the “no common shock” assumption is doubtful
 - ▶ With $\delta \approx 6.5$ the correction is large, and the test has low power
- ▣ The seminar repeats the test for 2020 and checks the sensitivity to the windows

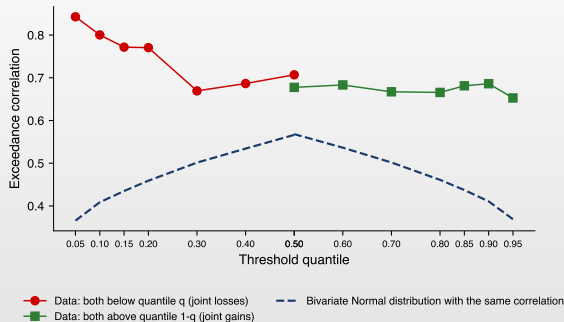
Contagion Tests beyond Forbes–Rigobon

- Corsetti, Pericoli & Sbracia (2005): with a common factor the Forbes–Rigobon adjustment over-corrects, biasing the test towards “no contagion”; they give a factor-model version
- Rigobon (2003): identification through heteroskedasticity; the variance shift between regimes identifies the structural coefficient β , an instrumental-variables idea
- δ is estimated, so ρ^* inherits the error of the variance ratio: resample the calm and crisis windows (stationary bootstrap, mean block 10 days, $B = 1999$)
 - ▶ 2008: $\hat{\delta} = 6.54$, 90% interval [3.9, 9.6]
 - ▶ BET: $\rho^* = 0.20$, 90% interval [0.16, 0.24], bootstrap p for $H_0 : \rho^* \leq \rho_{calm}$: 0.62
 - ▶ Euro Stoxx 50: $\rho^* = 0.44$ [0.38, 0.52], $p = 0.98$
- The 2008 verdict survives once the uncertainty of δ is included; the common-factor critique remains (Seminar A3, B4)

Exceedance Correlations

- ▣ Longin & Solnik (2001): correlation conditional on both returns being beyond a threshold
- ▣ Lower tail at quantile q : $\rho^-(q) = \text{Corr}(x, y \mid x \leq F_x^{-1}(q), y \leq F_y^{-1}(q))$
- ▣ Upper tail: the same with $x \geq F_x^{-1}(1 - q), y \geq F_y^{-1}(1 - q)$
- ▣ Benchmark: for a bivariate Normal distribution, $\rho^\pm(q) \rightarrow 0$ as $q \rightarrow 0$
 - ▶ Conditioning on extremes lowers the correlation even when nothing changes: compare with the Normal curve, not with ρ

Exceedance Correlations: S&P 500 and Euro Stoxx 50



- Weekly returns 2000–2026 ($n = 1\,393$, overall $\hat{\rho} = 0.78$); dashed: the bivariate Normal distribution with the same correlation (2 million simulated pairs)

Interpreting the Exceedance Chart

- ▣ At $q = 0.10$: joint losses $\rho^- = 0.80$ (86 weeks), joint gains $\rho^+ = 0.69$ (70 weeks); Normal benchmark 0.41
- ▣ Both tails are far above the Normal benchmark: extremes are more dependent than a Gaussian model allows
- ▣ The lower tail is higher than the upper tail: crashes are shared more than rallies
- ▣ A correlation coefficient cannot describe this: we need a model for the whole dependence structure $\Rightarrow$ copulas

Section Recap

- ▣ Higher volatility of the source market raises the measured correlation even with an unchanged mechanism
- ▣ Adjusted correlation: $\rho^* = \rho_c / \sqrt{1 + \delta(1 - \rho_c^2)}$
- ▣ 2008: raw increases significant, adjusted increases not; the result rests on strong assumptions
- ▣ Exceedance correlations are asymmetric and far above the Normal benchmark

Why Correlation Is Not Enough

- ▣ Embrechts, McNeil & Straumann (2002): three fallacies about linear correlation
- ▣ Correlation measures only *linear* dependence and is not invariant to monotone transformations:
 $\text{Corr}(x, y) \neq \text{Corr}(e^x, e^y)$
- ▣ Margins and correlation do not determine the joint distribution
 - ▶ Two portfolios with identical margins and ρ can have very different Value-at-Risk (Chapter 7)
- ▣ Attainable correlations: for given margins, $\rho \in [\rho_{\min}, \rho_{\max}]$, reached by countermonotone and comonotone pairs (Fréchet–Hoeffding bounds, next slides)
 - ▶ Lognormal $X = e^{\sigma_1 Z}$, $Y = e^{\pm \sigma_2 Z}$: $\rho_{\max} = \frac{e^{\sigma_1 \sigma_2} - 1}{\sqrt{(e^{\sigma_1^2} - 1)(e^{\sigma_2^2} - 1)}}$, $\rho_{\min}$ with $e^{-\sigma_1 \sigma_2}$ in the numerator
 - ▶ $\sigma_1 = 1$, $\sigma_2 = 3$: $\rho_{\max} = 0.162$ and $\rho_{\min} = -0.008$: perfectly dependent, yet almost uncorrelated

Sklar's Theorem



Main Building, Illinois Institute of Technology, Chicago: Abe Sklar (1925–2020) was professor of applied mathematics at IIT; no freely licensed portrait of him exists

Photo: Joe Ravi (2010); CC BY-SA 3.0; Wikimedia

[Commons](#)

- A **copula** is a joint distribution function on $[0, 1]^d$ with uniform margins
- **Sklar (1959)**: for any joint distribution F with margins $F_1, \dots, F_d$ there is a copula C with
 - ▶ $F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d))$
 - ▶ C is unique if the margins are continuous
- Conversely, any copula and any margins give a valid joint distribution
- If the margins and the copula have densities:
 $f(x) = c(F_1(x_1), \dots, F_d(x_d)) \prod_i f_i(x_i)$ (the comonotone copula has none)

What the Copula Separates

- ▣ Margins: volatility, skewness, fat tails of each asset (Chapters 1 and 5)
- ▣ Copula: all the dependence, and nothing else; invariant to increasing transformations of each variable
- ▣ Three reference copulas in two dimensions
 - ▶ Independence: $\Pi(u, v) = uv$
 - ▶ Comonotonicity (perfect positive dependence): $M(u, v) = \min(u, v)$
 - ▶ Countermonotonicity: $W(u, v) = \max(u + v - 1, 0)$
- ▣ Fréchet–Hoeffding bounds: $W \leq C \leq M$ for every copula
- ▣ Reference text: [Nelsen \(2006\)](#)

Rank Correlations: Kendall and Spearman

- **Kendall's tau** (Kendall (1938)): concordant minus discordant pairs
 - ▶ $\tau = P[(X_1 - X_2)(Y_1 - Y_2) > 0] - P[(X_1 - X_2)(Y_1 - Y_2) < 0]$
 - ▶ Copula form: $\tau = 4 \iint C \, dC - 1$
- **Spearman's rho** (Spearman (1904)): Pearson correlation of the ranks
 - ▶ Copula form: $\rho_S = 12 \iint (C(u, v) - uv) \, du \, dv$
- Both depend only on the copula; both equal 1 for M , 0 for Π , -1 for W

Links to copula parameters

Copula	τ
Gaussian, t	$\frac{2}{\pi} \arcsin \rho$
Clayton	$\theta / (\theta + 2)$
Gumbel	$1 - 1/\theta$
Frank	$1 - \frac{4}{\theta} [1 - D_1(\theta)]$

- $D_1(\theta) = \frac{1}{\theta} \int_0^\theta \frac{t}{e^t - 1} dt$ (Debye function)
- Gaussian: $\rho_S = \frac{6}{\pi} \arcsin(\rho/2)$

Elliptical Copulas: Gaussian and Student t

- **Gaussian:** $C_{\rho}^{Ga}(u, v) = \Phi_{2,\rho}(\Phi^{-1}(u), \Phi^{-1}(v))$
 - ▶ Density: $c = \frac{1}{\sqrt{1-\rho^2}} \exp\left(-\frac{\rho^2(x^2+y^2)-2\rho xy}{2(1-\rho^2)}\right)$, $x = \Phi^{-1}(u)$, $y = \Phi^{-1}(v)$
 - ▶ Symmetric, no tail dependence for $|\rho| < 1$
- **Student t** (Demarta & McNeil (2005)): $C_{\rho,\nu}^t(u, v) = t_{2,\rho,\nu}(t_{\nu}^{-1}(u), t_{\nu}^{-1}(v))$
 - ▶ ν degrees of freedom: small ν = strong joint extremes; $\nu \rightarrow \infty$ gives the Gaussian copula
 - ▶ Symmetric tail dependence, equal in both tails
- Easy to extend to d dimensions with a correlation matrix: the standard choice in risk systems

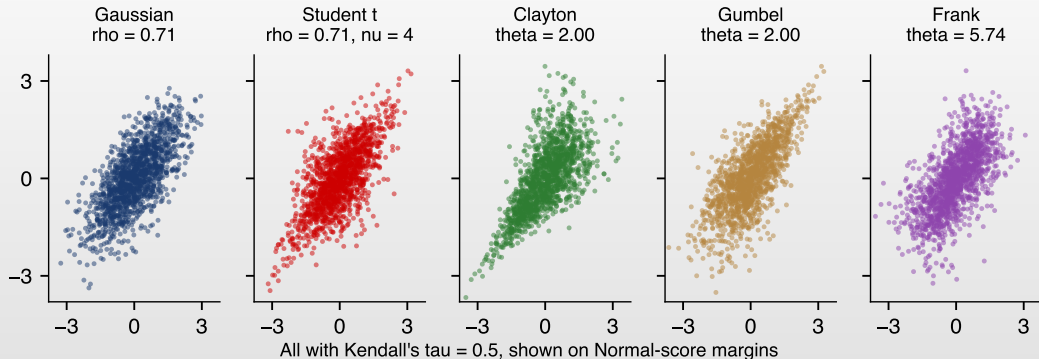
Archimedean Copulas: Clayton, Gumbel, Frank

- Built from a generator φ : $C(u, v) = \varphi^{-1}(\varphi(u) + \varphi(v))$
- **Clayton** (Clayton (1978)), $\theta > 0$
 - ▶ $C = (u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}$
 - ▶ lower-tail dependence only
- **Gumbel** (Gumbel (1960)), $\theta \geq 1$
 - ▶ $C = \exp(-[(-\ln u)^\theta + (-\ln v)^\theta]^{1/\theta})$
 - ▶ upper-tail dependence only
- **Frank** (Frank (1979)), $\theta \neq 0$
 - ▶ $C = -\frac{1}{\theta} \ln\left(1 + \frac{(e^{-\theta u} - 1)(e^{-\theta v} - 1)}{e^{-\theta} - 1}\right)$
 - ▶ symmetric, no tail dependence

Why use them

- One parameter, closed-form C , easy simulation
- Asymmetric tails: Clayton for joint crashes of returns, Gumbel for joint extremes of losses
- Rotating a copula (using $1 - u, 1 - v$) moves its tail dependence to the other corner
- Limit: exchangeable, poor in high dimension (one parameter for all pairs)

Five Copulas with the Same Kendall's Tau



- 1500 simulated pairs from each copula with $\tau = 0.5$, plotted on Normal-score margins $(\Phi^{-1}(u), \Phi^{-1}(v))$
- Parameters from the τ table: Gaussian and t $\rho = 0.707$, Clayton and Gumbel $\theta = 2.0$, Frank $\theta = 5.74$

Interpreting the Copula Zoo

- ▣ Same Kendall's tau, same margins: the clouds differ only in the corners
- ▣ Gaussian and Frank: the corners thin out, so joint extremes are rare
- ▣ Student t: both corners stay dense
- ▣ Clayton: a sharp lower-left corner (joint crashes); Gumbel: a sharp upper-right corner
- ▣ A risk manager who calibrates only τ or ρ can pick any of the five and get very different tail risk

Tail Dependence

- Lower tail:
 $\lambda_L = \lim_{q \downarrow 0} P(V \leq q \mid U \leq q) = \lim_{q \downarrow 0} C(q, q)/q$
- Upper tail: $\lambda_U = \lim_{q \downarrow 0} P(V > 1 - q \mid U > 1 - q)$
- $\lambda > 0$: extremes arrive together no matter how extreme;
 $\lambda = 0$: asymptotic independence
- Tail dependence is a copula property, not visible in ρ

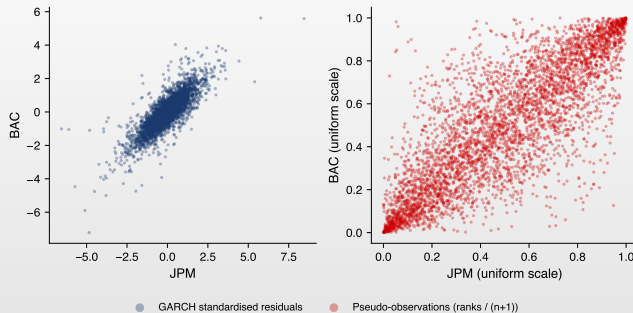
Closed forms

	λ_L	λ_U
Gaussian	0	0
Student t	$2 t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right)$	same
Clayton	$2^{-1/\theta}$	0
Gumbel	0	$2 - 2^{1/\theta}$
Frank	0	0

Estimating Copulas

- ▣ **Full maximum likelihood**: margins and copula together; efficient but heavy and fragile
- ▣ **IFM** (inference functions for margins, [Joe \(2005\)](#)): two steps
 - ▶ Fit parametric margins (e.g. GARCH with Student t); set $\hat{u}_{it} = \hat{F}_i(x_{it})$; maximise $\sum_t \ln c(\hat{u}_{1t}, \hat{u}_{2t}; \theta)$
- ▣ **CML** (canonical maximum likelihood, [Genest, Ghoudi & Rivest \(1995\)](#)): no parametric margins
 - ▶ Pseudo-observations $\hat{u}_{it} = \text{rank}(x_{it})/(T + 1)$; then maximise the copula likelihood
 - ▶ Robust to misspecified margins; the $T + 1$ keeps $\hat{u}$ away from 0 and 1
- ▣ For returns: filter each series with GARCH first, then use the ranks of the standardised residuals
- ▣ Model choice: $\text{AIC} = 2k - 2\ell$, $\text{BIC} = k \ln T - 2\ell$

From Residuals to Pseudo-Observations: JPM and BAC



- JPMorgan Chase (JPM) and Bank of America (BAC), daily 2010–2026 ($T = 4\,202$), GARCH(1,1)-filtered
- Kendall's $\hat{\tau} = 0.641$, Spearman's $\hat{\rho}_S = 0.822$; both corners of the uniform scatter are dense

Copula Estimates for JPM and BAC

Copula	$\hat{\theta}$ or $\hat{\rho}$	$\hat{\nu}$	ℓ	ΔAIC	λ_L	λ_U	fit p
Gaussian	0.833	–	2486.0	309.3	0.000	0.000	0.005
Student t	0.842	4.24	2641.7	0.0	0.531	0.531	0.010
Clayton	2.384	–	2116.7	1048.0	0.748	0.000	0.005
Gumbel	2.587	–	2412.2	456.9	0.000	0.693	0.005
Frank	9.152	–	2390.8	499.8	0.000	0.000	0.005

- CML on pseudo-observations of GARCH residuals; ΔAIC relative to the best model
- Student t wins by far: $\hat{\nu} = 4.2$ means strong joint extremes, $\hat{\lambda} = 0.53$ in both tails
- Gaussian beats Clayton, Gumbel and Frank: the dependence is strong and roughly symmetric
- Last column: goodness-of-fit p -values (next slides)

Rank Estimators and Their Precision

- ▣ $\hat{\tau}$ is a U-statistic; its Hoeffding projection gives $\sqrt{n}(\hat{\tau} - \tau) \rightarrow N(0, 16\text{Var}[2C(U, V) - U - V])$
- ▣ Plug-in: $\widehat{\text{se}}(\hat{\tau}) = 4 \text{sd}(2\hat{C}_n(\hat{U}_i, \hat{V}_i) - \hat{U}_i - \hat{V}_i) / \sqrt{n}$; inversion estimators $\hat{\theta} = g(\hat{\tau})$ by the delta method (Genest & Rivest (1993))
- ▣ Elliptical copulas: $\tau = \frac{2}{\pi} \arcsin \rho$ for every ν , so τ identifies ρ , never ν ; ν must come from the likelihood or from the tails
- ▣ JPM–BAC: $\hat{\tau} = 0.641$ (se 0.0065), so $\hat{\rho}_{\tau} = \sin(\pi\hat{\tau}/2) = 0.845$ (se 0.0055); CML: $\hat{\rho} = 0.842$ (se 0.0053, next slide)
- ▣ The moment estimator is almost as precise as CML for ρ and is a robust starting value for the likelihood

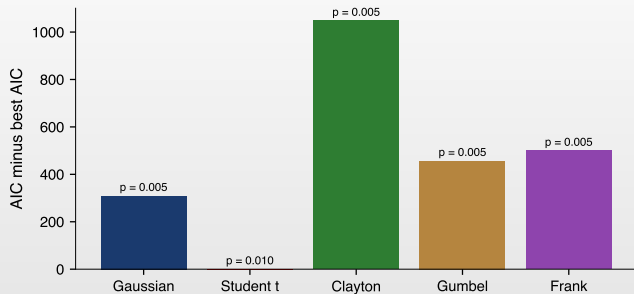
CML Standard Errors: The Ranks Are Estimated

- CML maximises $\sum_i \ln c(\hat{U}_i, \hat{V}_i; \theta)$ with *estimated* margins; the inverse Hessian assumes known margins and is too small
- Genest, Ghoudi & Rivest (1995): $\sqrt{n}(\hat{\theta} - \theta) \rightarrow N(0, A^{-1}\Sigma A^{-1})$, A the Fisher information of the copula, $\Sigma = \text{Var}[\dot{\ell}_\theta(U, V) + W_1(U) + W_2(V)]$
 - ▶ $W_1(u) = \iint 1\{u \leq u'\} \partial_u \dot{\ell}_\theta(u', v') dC(u', v')$: the price of ranking; W_2 likewise in v
 - ▶ Gaussian copula: variance $(1 - \rho^2)^2$ for CML, against $(1 - \rho^2)^2 / (1 + \rho^2)$ with known margins
- S&P 500–Euro Stoxx 50 weekly, Gaussian copula: se 0.0086 (inverse Hessian) versus 0.0138 (sandwich), +60%
- JPM–BAC, t copula: ρ : 0.0046 versus 0.0053 (+17%); ν : 0.359 versus 0.364 (+1%): ν is driven by the tails, where ranks carry little extra error

Chen and Fan (2006): GARCH Filtering Does Not Change the Limit

- Model: $r_{it} = \mu_i + \sigma_{it}\varepsilon_{it}$ with GARCH σ_{it} ; innovations $(\varepsilon_{1t}, \varepsilon_{2t})$ i.i.d. with copula C_θ
- [Chen & Fan \(2006\)](#): CML on the ranks of the *estimated* residuals $\hat{\varepsilon}_{it}$ has the same limit distribution as CML on the ranks of the true ε_{it}
- Consequence: the [Genest, Ghoudi & Rivest \(1995\)](#) sandwich stays valid after GARCH filtering; unlike DCC, no step-1 correction is needed
- Monte Carlo: 500 samples, $T = 2\,000$, Gaussian copula $\rho = 0.8$, Student $t(5)$ innovations, GARCH as fitted to JPM and BAC
 - ▶ sd of $\hat{\rho}$: 0.0081 (true innovations) and 0.0082 (GARCH residuals); mean sandwich se 0.0081, mean inverse-Hessian se 0.0063
- Goodness of fit on residuals: [Rémillard \(2017\)](#) shows that rank-based tests keep their limit and the parametric bootstrap applies, which justifies the Rosenblatt test presented below

Model Comparison and Goodness of Fit



Numbers above bars: goodness-of-fit p-value (Rosenblatt Cramer-von Mises, parametric bootstrap)

- Bars: AIC distance to the best copula; numbers: bootstrap p -values of the goodness-of-fit test

Goodness of Fit: The Rosenblatt Test

- If $(U, V) \sim C_\theta$, then $E_1 = U$ and $E_2 = h(V | U) = \partial C_\theta(U, V) / \partial u$ are independent uniforms
- Statistic (Genest, Rémillard & Beaudoin (2009)): $S_n = n \int_{[0,1]^2} (D_n(e) - e_1 e_2)^2 de$, D_n the empirical distribution of (E_1, E_2)
- p -value by parametric bootstrap, because θ is estimated
 - ▶ Simulate n pairs from $C_{\hat{\theta}}$, convert to pseudo-observations, re-estimate, recompute $S_n^{(b)}$
 - ▶ $p = (1 + \#\{S_n^{(b)} \geq S_n\}) / (B + 1)$; here $B = 200$ (100 for Student t)
- Result: every family is rejected at 5%; Student t has by far the smallest statistic (0.082 versus 0.968 for the Gaussian)
- Why all rejected?
 - ▶ $T = 4202$ gives high power against small departures; one candidate departure: a static copula over 16 years ignores that dependence may move (Section 3)
 - ▶ Remedy: time-varying (conditional) copulas, Patton (2006); a dynamic t copula follows at the end of this section

Empirical Tail Dependence: JPM and BAC



- Estimates $\hat{\lambda}_L(q) = \hat{C}(q, q)/q$ and $\hat{\lambda}_U(q)$ for q from 1% to 20%, with the 90% band of the same estimator on data simulated from the fitted t copula

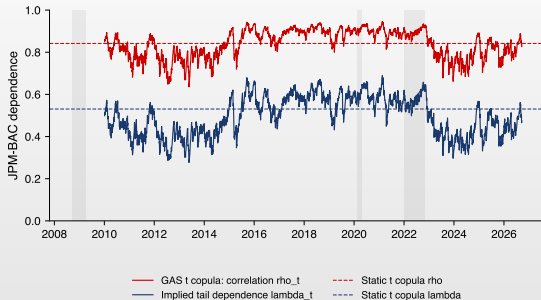
Interpreting the Tail-Dependence Chart

- ▣ At $q = 5\%$: $\hat{\lambda}_L = 0.61$, $\hat{\lambda}_U = 0.55$; at $q = 1\%$: 0.52 and 0.43
- ▣ When JPM has one of its 5% worst days, BAC is in its own worst 5% about 61% of the time
- ▣ The lower-tail estimates lie inside the t-copula band (0.57–0.65 at 5%): the t copula describes the crashes well
- ▣ The upper tail looks weaker (joint rallies less tight than joint crashes); the next slide tests whether this asymmetry is more than noise
- ▣ Finite- q estimates are biased and noisy: λ is a limit, and very few points lie in the far corner

Estimating Tail Dependence Nonparametrically

- Schmidt & Stadtmüller (2006): $\hat{\lambda}_L = \frac{n}{k} \hat{C}_n(\frac{k}{n}, \frac{k}{n})$, the chart estimator with $q = k/n$; $\hat{\lambda}_U$ likewise in the upper corner
- Consistent and asymptotically Normal if $k \rightarrow \infty$, $k/n \rightarrow 0$, the tail copula exists and a second-order bias condition holds ($\sqrt{k} A(n/k) \rightarrow 0$): a larger k lowers the variance but raises the bias
- Is the JPM–BAC asymmetry real at fixed q ? $H_0 : \lambda_L(q) = \lambda_U(q)$ (finite-threshold probabilities, not the limits); bootstrap of residual pairs, re-ranked, $B = 999$
 - ▶ $q = 5\%$ ($k = 210$): $\hat{\lambda}_L - \hat{\lambda}_U = 0.062$, 95% interval i.i.d. $[-0.014, 0.148]$, stationary block bootstrap $[-0.014, 0.138]$
 - ▶ $q = 1\%$ ($k = 42$): 0.095 , i.i.d. $[-0.095, 0.286]$, block $[-0.119, 0.286]$: wider, as the bias–variance trade-off predicts
 - ▶ The i.i.d. bootstrap assumes a static innovation copula; the block version (mean block by the Politis & White (2004) rule: 1.6 and 1.0 days) allows serial dependence in the tail indicators
- All intervals contain zero: at these thresholds the tail asymmetry is not significant, so the symmetric t copula is not contradicted in the tails

A Dynamic t Copula for JPM and BAC



- GAS (generalised autoregressive score) t copula, [Creal, Koopman & Lucas \(2013\)](#): $\rho_t = \tanh f_t$, $f_{t+1} = \omega + A s_t + B f_t$, $s_t = \partial \ln c(\hat{u}_t, \hat{v}_t; \rho_t, \nu) / \partial f_t$ (unit scaling)
- Same pseudo-observations as the static fit; implied $\lambda_t = 2t_{\nu+1} \left(- \sqrt{(\nu+1)(1-\rho_t)/(1+\rho_t)} \right)$

Interpreting the Dynamic Copula

- ▣ Estimates: $A = 0.025$, $B = 0.9926$, $\nu = 5.35$ (static 4.24); likelihood ratio against the static t copula 245.5
- ▣ ρ_t ranges from 0.64 (23 May 2013) to 0.95 (18 February 2021); λ_t from 0.28 to 0.69, around a static 0.53
- ▣ Once ρ moves, a larger ν suffices: part of the static fat tails was time variation in dependence
- ▣ Under $A = 0$, B is not identified (the Davies problem again): judge the likelihood ratio by bootstrap, not by χ^2 ; here the gap is very large
- ▣ Time-varying dependence is one explanation consistent with the rejection of every static family; rejection alone does not single it out (Seminar B6)

Dependence in High Dimension and over Time

- DCC for large N : the full likelihood is slow and biased
 - ▶ Composite likelihood over pairs of assets, [Pakel, Shephard, Sheppard & Engle \(2021\)](#); DCC-NL, nonlinear shrinkage of the target $\bar{Q}$ (the shrinkage of Chapter 4), [Engle, Ledoit & Wolf \(2019\)](#)
- Vine (pair-copula) constructions, [Aas, Czado, Frigessi & Bakken \(2009\)](#)
 - ▶ $d(d - 1)/2$ bivariate copulas, some conditional, in nested trees; each pair has its own family and tails
- Hierarchical Archimedean copulas, [Okhrin, Okhrin & Schmid \(2013\)](#)
 - ▶ Groups nested by the strength of dependence: banks within a country, countries within a region
- Dynamic copulas
 - ▶ Score-driven parameters ([Creal, Koopman & Lucas \(2013\)](#)) extend to many assets through factor copulas; [Oh & Patton \(2018\)](#) measure time-varying systemic risk in credit default swap spreads

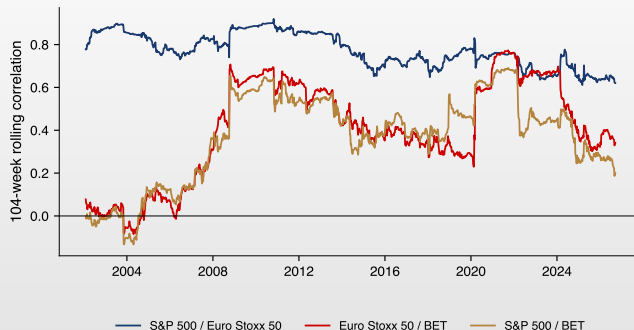
A Cautionary Tale: The Gaussian Copula and 2008

- ▣ Li (2000) modelled joint default times with a Gaussian copula
 - ▶ One correlation parameter, calibrated to market spreads, priced whole portfolios of credit
- ▣ It became the market standard for pricing tranches of collateralised debt obligations (CDO)
- ▣ What went wrong
 - ▶ Zero asymptotic tail dependence: for the most extreme joint defaults it gives lower probabilities than tail-dependent models calibrated to the same data
 - ▶ A static correlation calibrated in calm years: exactly the Forbes–Rigobon and DCC lessons
- ▣ The copula is a tool; the danger was a single, convenient, thin-tailed choice

Section Recap

- ▣ Sklar: joint distribution = margins + copula; the copula holds all the dependence
- ▣ Kendall's tau and Spearman's rho depend only on the copula and map one-to-one to one-parameter families
- ▣ Tail dependence: 0 for Gaussian and Frank, symmetric for t, lower for Clayton, upper for Gumbel
- ▣ Estimate on pseudo-observations of GARCH residuals; test fit with the Rosenblatt statistic and a parametric bootstrap
- ▣ CML standard errors need the rank term of Genest–Ghoudi–Rivest; after GARCH filtering nothing else changes (Chen–Fan)
- ▣ JPM–BAC: t copula best among static ones, but rejected; a GAS t copula with moving ρ_t fits far better

US, Euro-Area and Romanian Equities, 2000–2026

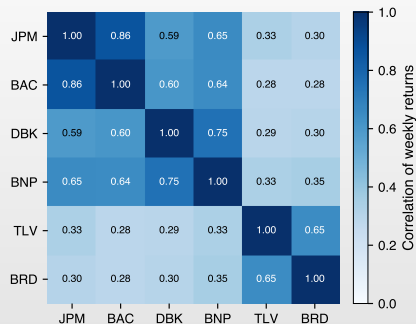


Correlations of weekly returns on a 104-week window (common trading days, Friday closes)

Interpreting Market Integration

- ▣ Euro Stoxx 50–BET: 0.33 in 2000–2009, 0.45 in 2010–2019, 0.61 since 2020
- ▣ S&P 500–Euro Stoxx 50: 0.84, 0.78, 0.70 in the same periods
- ▣ Rolling Euro Stoxx 50–BET correlation: 0.31 at end-2007, 0.65 at end-2009, 0.29 at end-2019, 0.76 at end-2021, 0.35 now
 - ▶ Jumps in crises, partial return afterwards: integration or the Forbes–Rigobon volatility bias? The seminar tests it
- ▣ Lower correlation means better diversification for euro-area investors holding Romanian equities

Banks: Correlations of Weekly Returns, 2010–2026

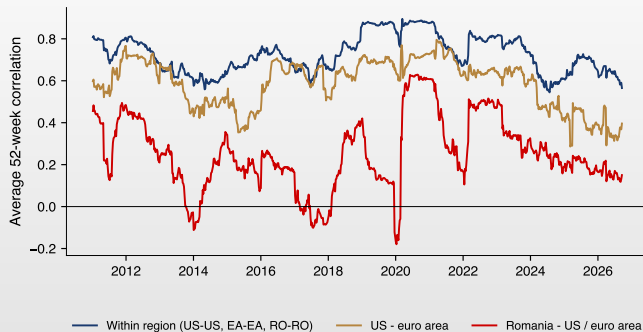


- US: JPM and BAC; euro area: DBK (Deutsche Bank) and BNP (BNP Paribas); Romania: TLV (Banca Transilvania) and BRD (BRD-Groupe Société Générale)
- 871 weeks from 15 January 2010

Interpreting the Bank Correlations

- ▣ Three blocks: US (0.86), euro area (0.75), Romania (0.65)
- ▣ Cross-region links are weaker for Romania: e.g. BNP–TLV 0.33, JPM–BRD 0.30
- ▣ BRD is controlled by Société Générale, yet its stock co-moves more with TLV than with any euro-area bank
- ▣ Bank returns share country-level factors: sovereign risk, domestic rates, regulation

Bank Correlations over Time

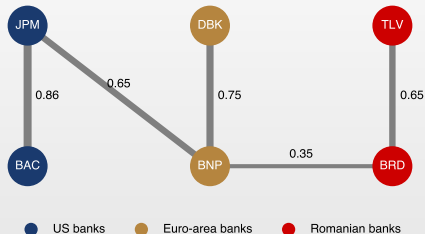


- Average of pairwise 52-week correlations of weekly returns within regions, between the US and the euro area, and between Romania and the others

Interpreting the Bank Correlations over Time

- ▣ Full-period averages: within regions 0.73, US–euro area 0.59, Romania–others 0.25
- ▣ March–June 2020: 0.87, 0.69 and 0.56: the Romanian link more than doubled in a global shock
- ▣ Latest values: 0.57, 0.40, 0.15
- ▣ Diversification across banking systems works in normal times and weakens exactly in systemic events

A Correlation Network of Banks



- Minimum spanning tree (Mantegna (1999)) on distances $d_{ij} = \sqrt{2(1 - \rho_{ij})}$: the 5 strongest links that connect all 6 banks without cycles
- Edges: JPM-BAC (0.86); JPM-BNP (0.65); DBK-BNP (0.75); BNP-BRD (0.35); TLV-BRD (0.65); Romania is attached through BNP-BRD

From Correlations to Systemic Risk

- ▣ The tree summarises the strongest co-movement links $US \leftrightarrow \text{euro area} \leftrightarrow \text{Romania}$; it is undirected and contemporaneous, so it does not identify shock-transmission paths, only hypotheses to test
- ▣ Correlation networks are the first step towards systemic-risk measures (Chapter 18)
 - ▶ Connectedness from Granger causality: [Billio, Getmansky, Lo & Pelizzon \(2012\)](#)
 - ▶ Spillover indices from variance decompositions: [Diebold & Yilmaz \(2009\)](#)
 - ▶ Tail-event networks: the Financial Risk Meter applied to EU energy firms, [Toma, Dima, Grecu, Pele & Lessmann \(2026\)](#)
- ▣ Energy prices and EU stock markets: dependence across regions and industrial groups, [Grecu, Cramer, Pele & Lessmann \(2025\)](#) (case study below)
- ▣ Correlation networks inherit all the problems of correlations: non-linearity, time variation, tails
- ▣ Distances: here $\sqrt{2(1-\rho)}$, in Chapter 4 (hierarchical risk parity) $\sqrt{(1-\rho)/2}$; they differ by a factor 2, both increase with $1-\rho$ and give the same tree

Section Recap

- ▣ BET–Euro Stoxx 50 weekly correlation rose from 0.33 (2000s) to 0.61 (since 2020)
- ▣ Bank returns cluster by region; Romanian banks are the least connected, except in global shocks
- ▣ Minimum spanning trees summarise a correlation matrix as a network, the bridge to Chapter 18

Research Question and Motivation

- [Grecu, Cramer, Pele & Lessmann \(2025\)](#): how do oil and gas prices co-move with, and shock, the main stock index of each EU member state?
 - ▶ Does the link differ by region and by degree of industrial development?
- Why it matters
 - ▶ The EU is a net energy importer; the 2022 invasion of Ukraine and the Green Deal reshaped its energy supply
 - ▶ Channels from energy to equities: stock valuation, monetary policy, output and uncertainty
- Gap addressed
 - ▶ Earlier work: mostly one country or a few developed markets, mostly crude oil
 - ▶ Here: all EU markets, natural gas next to oil, regional and industrial groupings
- Link to this chapter
 - ▶ The DCC model of Section 3 applied to energy–equity pairs; crisis spikes read with the Forbes–Rigobon lens of Section 4

Data and Method, as Published

- **Data**, 2010–2023: main stock index of each EU country; generic (rolled) futures on Brent crude oil and on European natural gas (Dutch TTF)
 - ▶ Daily returns for the DCC model; quarterly returns for the TVP-VAR, with global industrial production and oil production (annual growth rates) as controls
- **Groups**: four EuroVoc regions (Central and Eastern, Northern, Southern, Western Europe); most versus less industrialised countries (Eurostat shares of EU sold industrial production, 2022)
 - ▶ Group parameter $\alpha_R = \sum_{i \in R} \alpha_i \overline{MC}_i / \overline{MC}_R$, $\overline{MC}$ the average 2010–2023 market capitalisation (Eq. 1–2)
- **Model 1**: DCC-GARCH (Engle & Sheppard (2001)): $H_t = D_t R_t D_t$, univariate GARCH volatilities in D_t , correlation recursion in (a, b) as in Section 3 (Eq. 3–5)
 - ▶ One bivariate model per index–energy pair; output: the path of ρ_t
- **Model 2**: TVP-VAR (Primiceri (2005)): $y_t = X_t \beta_t + \varepsilon_t$, $\beta_t = \beta_{t-1} + v_t$, Normal innovations (Eq. 6–7)
 - ▶ Bayesian estimation by Gibbs sampling; generalised impulse responses (GIRF) with 95% bands

Key Results

	Oil max	Oil min	Gas max	Gas min
Central & Eastern	0.53 (2020)	−0.11 (2022)	0.23 (2015)	−0.43 (2022)
Northern	0.55 (2020)	−0.15 (2022)	0.37 (2015)	−0.35 (2014)
Southern	0.64 (2020)	−0.21 (2022)	0.32 (2015)	−0.34 (2022)
Western	0.67 (2020)	−0.36 (2022)	0.33 (2015)	−0.35 (2022)
Most industrialised	0.67 (2020)	−0.33 (2022)	0.34 (2015)	−0.35 (2022)
Less industrialised	0.57 (2020)	−0.20 (2022)	0.35 (2015)	−0.35 (2022)

Extreme DCC correlations of the group indices, [working paper \(2024\)](#), Tables 1–3

- Oil: peak in 2020 everywhere, a common COVID-19 shock to both markets; trough after the 2022 invasion
- Western and Southern markets co-move most with oil; industrialised markets more than the rest (0.67 vs 0.57)
- Gas: weaker link; lowest point −0.43 in Central and Eastern Europe in 2022, closest to the war; no industrial gap
- TVP-VAR, one-s.d. gas shock (Table 4): −0.78 pp at horizon 2 (5% significant), cumulative −0.93 pp (industrialised) vs −0.68 pp: stronger where industry weighs more
- Published version: a Markov-switching model finds high- and low-correlation regimes, each persisting with probability of about 98%

Replicate, Limits and an Open Extension

- **Replicate (oil leg):** reuse the DCC code of Section 3 ([Quantlet](#)) on DAX, WIG20, PX, BUX, BET (data/market) paired with Brent
 - ▶ Brent spot price (Europe), U.S. Energy Information Administration, via FRED; join prices on common days, then returns, 2010–2023
 - ▶ Check: the 2020 peak and the 2022 trough reported for the regional indices (Table 1)
- **Data limits**
 - ▶ Course data have spot Brent, not generic futures, and no daily TTF series: the gas leg cannot be replicated exactly
- **Scope of the tools**
 - ▶ DCC correlations are linear, symmetric and contemporaneous; direction and timing come from the TVP-VAR block
- **Open extension:** is the post-2022 energy–equity link a tail phenomenon?
 - ▶ Fit the copulas of Section 5 to DCC-GARCH residuals before and after February 2022; compare lower-tail dependence λ_L with the DCC trough, with bootstrap intervals

An Open Question

- Is the positive stock–bond correlation since 2022 a durable regime governed by inflation, and can it be forecast?
- The question is live
 - ▶ Campbell, Pflueger & Viceira (2020): the sign of the bond–stock comovement follows the inflation–output relation and the monetary-policy regime
 - ▶ Molenaar, Sénéchal, Swinkels & Wang (2024): the correlation across countries and a long history, and its links to macroeconomic conditions such as inflation
 - ▶ This chapter: daily SPY–TLT correlation -0.40 in 2002–2021 and 0.11 in 2022–2026; DCC above its target on 92% of days since 2022
- Formal statement with monthly realised correlations ρ_m
 - ▶ $z_m = \operatorname{artanh} \rho_m = a + b \pi_{m-2} + u_m$, π the annual US CPI inflation known when month m starts
 - ▶ H_0 : $b = 0$; H_1 : $b > 0$; HAC t-statistic; falsified if $b \approx 0$ in periods where inflation varied but the regime did not

The Discovery Loop, Step by Step (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List studies since 2015 on the drivers of the stock–bond correlation, with DOI, sample, measure of correlation, identification strategy and main estimate."*
 - ▶ Resolve every DOI on Crossref and check the estimate in the paper, not in the summary
- **Hypotheses:** *"Give competing explanations for the 2022 sign change (inflation level, monetary-policy regime, term-premium shock, a one-off tightening) and what each predicts for ρ before 2022 and out of sample."*
- **Data and code:** *"Write Python that joins SPY and TLT adjusted prices from data/market on common days, computes monthly realised correlations and merges US CPI inflation published before each month starts."*
 - ▶ Replicate first: -0.40 (2002–2021) and 0.11 (2022–2026), the numbers of Section 1

The Discovery Loop, Step by Step (2/2)

- **Robustness:** *“Write a pre-registered plan: the pre-2022 subsample, core CPI instead of headline, weekly instead of daily returns, DCC instead of realised correlation, control for realised volatility, HAC lag 12.”*
 - ▶ Placebo: replace π by simulated random walks of the same persistence; a persistent regressor can produce large t-statistics by chance
- **Critique:** *“Act as a referee: is b identified by a single episode, and would the Forbes–Rigobon volatility bias explain it? Give the test that would expose each.”*
- **Report:** every inflation measure, window and frequency tried; subsample estimates next to the full sample
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

What the Researcher Must Check

- **Look-ahead:** CPI for month m is published during month $m + 1 \Rightarrow$ use π_{m-2} and check the release calendar
 - ▶ The lag handles publication delay, not revisions: the mini-case uses the current vintage of the seasonally adjusted CPI, revised every year; real-time claims need historical vintages or the unrevised non-seasonally-adjusted CPI
- **Calendars:** returns computed before joining create spurious pairs $\Rightarrow$ join prices on common days first (Section 2)
- **Persistent regressor, few regimes:** inflation is near a unit root and changed regime once $\Rightarrow$ subsample estimates, long HAC lags, a placebo with simulated persistent series
- **Volatility bias:** measured correlation moves with volatility (Section 4) $\Rightarrow$ add realised volatility as a control
- **Multiple testing:** several inflation measures, windows and frequencies $\Rightarrow$ fix them in the pre-registration
- **Invented references:** an unresolvable DOI or a misquoted estimate $\Rightarrow$ drop the claim and log it in `AI_ERRORS.md`

Mini-Case: Does Inflation Drive the Stock–Bond Correlation?

- A typical assistant proposal: “high inflation makes the stock–bond correlation positive”; $H_0: b = 0$
- Replicate: daily SPY–TLT correlation -0.40 (2002–2021) and 0.11 (2022–2026)
- Full sample, 287 months (September 2002 – August 2026): $\hat{b} = 0.055$ per percentage point of inflation, HAC s.e. 0.019 , $t = 2.96$, $R^2 = 0.047$
 - ▶ Mean monthly ρ : -0.12 when inflation exceeds 3% (89 months), -0.30 otherwise (198)
- Identification check, 2002–2021 only (232 months, 53 of them with inflation above 3%): $\hat{b} = 0.002$, $t = 0.06$
- Interpretation (associational): the full-sample slope is sensitive to the post-2022 episode; the pre-2022 slope is not distinguishable from zero, which is not proof of no effect; a regime difference needs an interaction test
- One regime change cannot separate “inflation level” from “policy regime” (Campbell, Pflueger & Viceira (2020)): the claim is not yet established

Your Turn: A Team-Project Seed

- Question: does a DCC model whose correlation target depends on inflation and the policy rate forecast the SPY–TLT correlation better out of sample than the plain DCC?
- Replicate first: the DCC estimates of Section 3 ($a + b = 0.986$, $\bar{Q}_{12} = -0.25$) and the mini-case slope $\hat{b} = 0.055$
- Deliverables
 - ▶ A one-page pre-registration: hypotheses, forecast loss, subsamples, placebo and what would falsify the claim
 - ▶ Code on data/market plus the official inflation series, a Diebold–Mariano comparison, a referee-style self-critique
- Log every prompt, tool and accepted output in `AI_USE.md`; every error caught (look-ahead in CPI, returns before joining, invented DOI) and the test that caught it in `AI_ERRORS.md`
- Code of the mini-case: [Quantlets/Ch_06/ai_discovery_case.py](#); inflation: US consumer price index for all urban consumers, U.S. Bureau of Labor Statistics

Common Pitfalls

- ▣ Computing returns before joining calendars, or comparing daily correlations across time zones
- ▣ Testing a break at a date chosen by looking at the chart: use an unknown-date test
- ▣ Calling a crisis increase “contagion” without the Forbes–Rigobon adjustment
- ▣ Reading copula fit only from AIC: the best of five can still be rejected
- ▣ Assuming a Gaussian copula for risk: zero tail dependence by construction
- ▣ Trusting step-2 standard errors, inverse-Hessian copula standard errors or Fisher intervals under volatility clustering

Where This Chapter Fits

Builds on

- ▣ Chapter 1: fat tails and volatility clustering of single series
- ▣ Chapter 4: portfolio optimisation needs Σ
- ▣ Chapter 5: univariate GARCH is step 1 of DCC and the filter before copulas

Leads to

- ▣ Chapter 7: portfolio Value-at-Risk and Expected Shortfall with DCC covariances and simulated copulas
- ▣ Chapter 8: backtesting multivariate risk forecasts
- ▣ Chapter 18: systemic risk and financial networks

Conclusions

- ▣ Correlations move: SPY–TLT changed sign, Bitcoin–SPY rose from about zero to 0.31, BET joined the European cycle
- ▣ EWMA is simple, DCC estimates the dynamics with $3N + 2$ parameters and rejects constant correlation
- ▣ Crisis correlations must be adjusted for volatility before speaking of contagion
- ▣ Copulas separate margins from dependence; tail dependence is what matters for joint losses
- ▣ For JPM–BAC every static copula is rejected and a dynamic t copula fits far better: evidence that dependence moves over time

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Measuring Statistical Risk
- ▣  Multivariate Statistical Analysis
- ▣  Minimum Spanning Tree
- ▣  FRM: Financial Risk Meter

Code, data and textbooks

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_06
- ▣ Textbooks: Franke, Härdle & Hafner (2019); Härdle & Simar (2019); Nelsen (2006)

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